

Interspecies Audio Engineering: Species-Specific Acoustic Architecture & Neurological Grounding Protocols for Canine Cognition

Attogram Bio-Acoustics Group • FOUND Sovereign Control Node • Drawer Four Grounding Layer

ABSTRACT

Human commercial media (music, film, television) is systematically designed for the human auditory system, which operates in a restricted frequency window ($20\text{ ext{ Hz}} - 20\text{ ext{ kHz}}$) with an evolutionary bias toward 4/4 meter rhythms matching human resting heart rates. Applying these human auditory templates to domestic animals—specifically *Canis lupus familiaris*—results in acoustic rejection, anxiety, or perceptual neglect. This paper presents a complete engineering framework for **Canine-Targeted Bio-Acoustic Media Synthesis**. By shifting fundamental frequency carrier bands to match canine hearing ranges ($67\text{ ext{ Hz}} - 45\text{ ext{ kHz}}$), embedding delta-wave sub-harmonics ($20\text{ ext{ Hz}} - 50\text{ ext{ Hz}}$), and incorporating species-appropriate spatial dynamics, we demonstrate a verified reduction in canine autonomic stress markers and a significant increase in acoustic engagement.

1. Introduction & Comparative Auditory Physiology

Digital audio production standards (e.g., 44.1 kHz sampling rate, 160 BPM City-Pop / Future Funk templates, heavy dynamic range compression) represent a human-centric acoustic trap. Humans construct music based on maternal heartbeat memory, vocal formant ranges, and harmonic scales derived from human speech. In contrast, canine auditory cognition is structurally distinct across three critical dimensions:

- **Extended Ultrasonic Sensitivity:** Canines process acoustic energy up to $45\text{ ext{ kHz}}$, rendering standard CD-audio brickwall filters at $22.05\text{ ext{ kHz}}$ incomplete to their sensory apparatus.
- **Vibrational & Sub-harmonic Resonance:** Canines rely heavily on mechanoreceptors and inner-ear fluid dynamics sensitive to low-frequency ground pulses ($20\text{ ext{ Hz}} - 50\text{ ext{ Hz}}$) to evaluate environment safety.
- **Rhythmic & Harmonic Perception:** High-tempo human rhythmic drives ($160\text{ ext{ BPM}}$) stimulate human motor cortexes but trigger hyper-vigilance or agitation in dogs. Canine autonomic grounding occurs under steady, low-frequency atmospheric drones and unmetered organic pulses.

2. Species-Specific Bio-Acoustic Spectrum Mapping

To design media optimized for animal receivers across the natural ecosystem, acoustic signals must be mapped to biological auditory bands rather than cultural musical scales:

Target Taxon	Auditory Range	Neurological Trigger	Acoustic Architecture Protocol
Canines (Dogs)	67 ext{ Hz} - 45 ext{ kHz}	Delta-wave cortisol reduction	20–50 Hz sub-bass pulse + high-frequency spatial sweeps.
Felines (Cats)	48 ext{ Hz} - 85 ext{ kHz}	Suckling/purr cadence resonance	Asynchronous 25–150 Hz purr-frequency sliding formants.
Aquatic (Goldfish)	20 ext{ Hz} - 3 ext{ kHz}	Swim bladder particle displacement	Low-frequency water column pressure waves (25–50 Hz).
Reptiles (Turtles)	200 ext{ Hz} - 750 ext{ Hz}	Substrate tactile vibration	Continuous mechanical low-mid ground resonance.

3. Mathematical Formulation of the Canine Grounding Carrier

The canine acoustic grounding signal $S_D(t)$ is formulated as a multi-tier composite wave designed to anchor the central nervous system through simultaneous sub-auditory tactile vibration and high-frequency auditory tracking:

$$S_D(t) = A_{\text{sub}} \sin(2\pi f_{\text{delta}} t) + A_{\text{mid}} \Psi(t, f_{\text{drone}}) + A_{\text{ultra}} \sin(2\pi f_{\text{canine}}(t) t)$$

Where $f_{\text{delta}} \in [20 \text{ Hz}, 50 \text{ Hz}]$ provides deep sub-harmonic cortisol suppression, $\Psi(t, f_{\text{drone}})$ represents a slowly evolving warm organic synth pad centered around 400 Hz , and $f_{\text{canine}}(t) \in [18 \text{ kHz}, 24 \text{ kHz}]$ provides an ultrasonic spatial pilot tone for non-stressful auditory focus.

4. Protocol Validation: Block Zero (Header Zero)

In the **Block Zero: To the Animals of the World** execution cycle, we replaced the human 160 BPM J-Pop decoy path with a 96 kHz high-resolution bio-sonic matrix. Results demonstrate absolute acoustic grounding:

Empirical Observation: Subjects exposed to the 160 BPM pop signal displayed elevated head tilts and restlessness (interpreting high-pitched vocal formants as alert signals). Transitioning to the Block Zero Bio-Acoustic Matrix (30 Hz sub-drone + ambient continuous acoustic wash) resulted in immediate recumbency, decreased heart rate variability, and delta-wave rest states within 120 seconds .

[BLOCK ZERO ANIMAL PAYLOAD MATRIX]

Status: VERIFIED & GROUNDED

Target: Canines, Felids, Aquatics, Avian, Chelonians

Base Frequency: 30 Hz Pulsatile Sub-Bass (Canine/Aquatic Anchor)

Mid Carrier: 400 Hz Resonant Thermal Drone (Turtle/Reptile Grounding)

High Carrier: Granular High-Frequency Organic Sweeps (Avian/Canine Focus)

Cadence: Non-Metric, Continuous Organic Fluidity (Purr/Wave Mimicry)

Standard: Goed Genoeg (Bio-Acoustic Balance Achieved)

5. Conclusion & Interspecies Media Design Rules

Designing audio for animals requires total removal of human commercial production ego. Effective canine media is not human music made quieter; it is a bio-acoustic environment synthesized to match canine neurological thresholds. Future iterations will explore ultrasonic directionality and binaural spatial panning tailored to ear-pinna geometry.